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SLEEP DIARY STUDY

Background

A modern day college student has one of the most irregular sleep schedules due to academic demands, social pressures, and lifestyle changes. By conducting a study like this we can find key insights to see how sleep affects a college student's performance in order to help universities design more effective wellness programs and inform interventions aimed at improving student health.

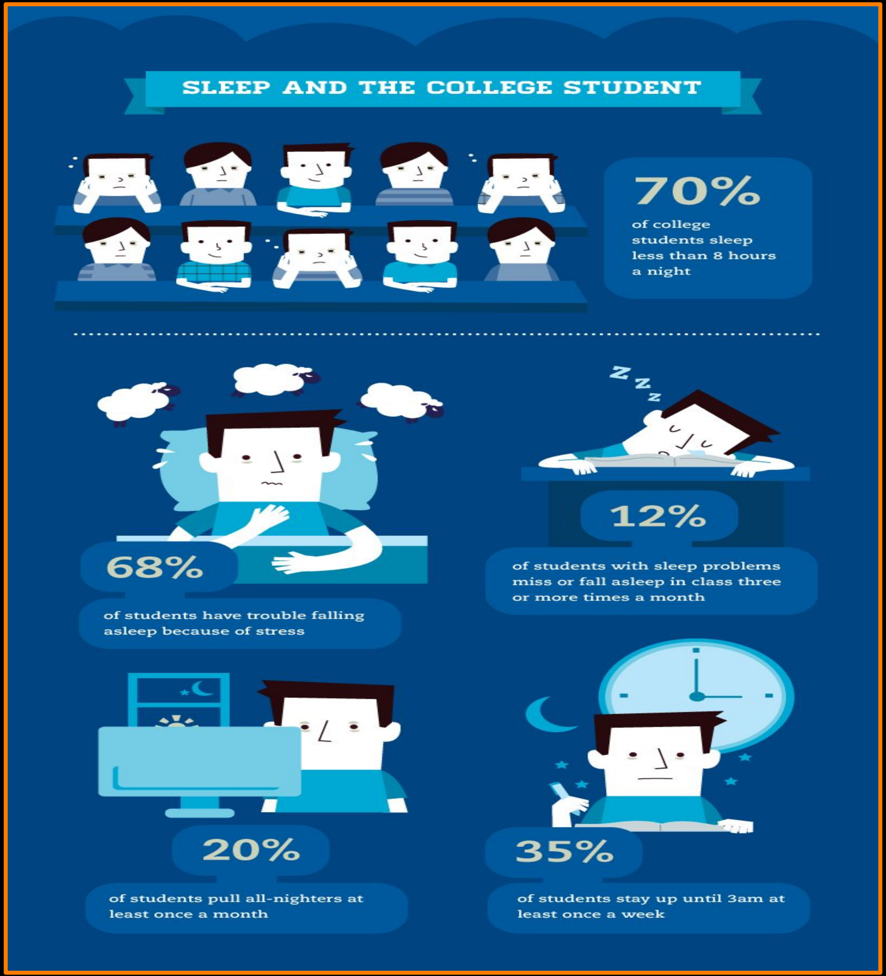

Image Credit: slugbooks.com

Motivation

Poor sleep habits have been linked to declines in cognitive performance, mood regulation, physical health, and overall academic success. To the right is a graphic detailing the negative effects of poor sleep.

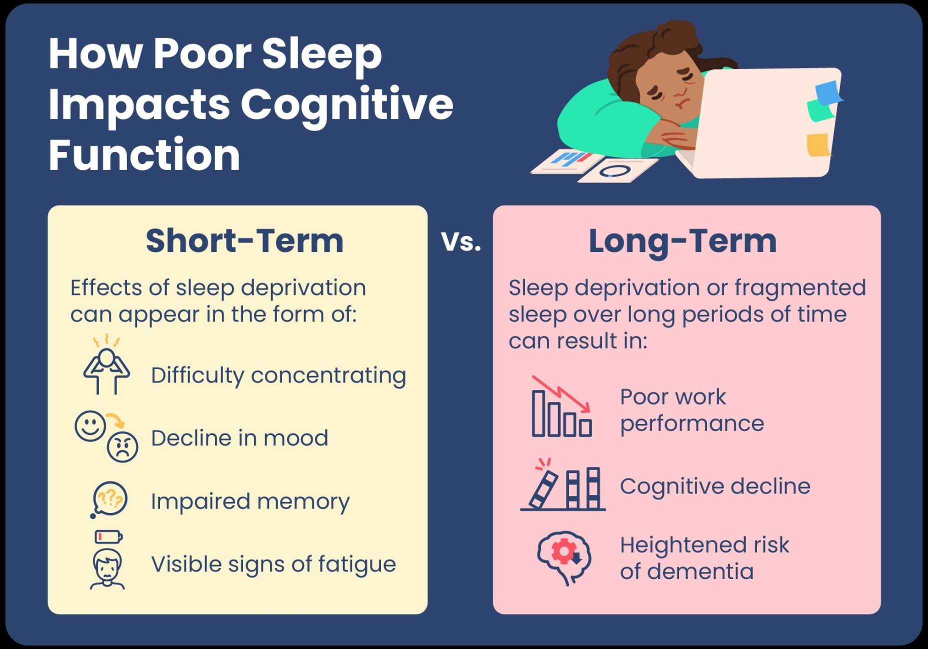

Image Credit: sleepfoundation.org

Research Objectives

★ Through this study, we aim to get a better understanding as to how nighttime routines affect the quality of sleep that people get
★ Identify any patterns there might be among participants’ nighttime routines
★ Explore how these night routines affect their emotional state and their alertness in the morning
★ Get insights into how we can create solutions that support healthier nighttime routines to help improve people's everyday lives

Study Design

Research Method

★ Diary Study

Study Duration

★ 4 days per participant

Why Diary Study?

★ Allows us to capture real-life behaviors over a period of time and allows us to easily sort through the data to recognize any patterns
★ In this case, it allows both nighttime and morning reflections to get a more detailed analysis

Participants & Recruitment

Participants

- ★ 4 participants total (1 participant per group member)
- ★ All participants being college students

Recruitment Process

- ★ Reach out to people we knew from our personal academic and social networks

General Demographics

- ★ Age: 18-22
- ★ Gender: Both males and females
- ★ Ethnicity: Mixed backgrounds

Data Collection

Data Collection Process

★ Participants were asked to complete nightly diary entries before going to bed
★ Completed morning reflections diary entry the next morning after waking up

When

★ Everyday from Wednesday - Sunday, 4 consecutive days

Where

★ Digital Diary on google forms

Types of Data Collected

Qualitative Data which included:

★ Activities included in nighttime routine
★ Emotional state before and after completing nighttime routine
★ Morning reflections on quality of sleep
★ Morning mood, alertness, and preparedness for the day ahead

Quantitative Data which included:

★ Bedtime and wake time
★ Sleep duration
★ Ratings of sleep quality
★ Mood and alertness ratings

CLUSTERS

Sleep Quality

Short sleep (5–6 hours) → fatigue, low mood

Anabhayan

Longer sleep (7–9 hours) → energized, productive

Anabhayan

Irregular bedtimes / late-night studying → anxiety, morning regret

Anabhayan

Mood Changes

Mood changes: drained → balanced

Anabhayan

Motivation / productivity rises with consistent sleep

Anabhayan

Anxiety appears with short or irregular sleep

Anabhayan

Night Routines

Hygiene routines (skincare, brushing teeth)

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Reading, journaling, meditation

Anabhayan

Family time / social interaction

Anabhayan

Screen time before bed (phone, TV, YouTube)

Anabhayan

Sleep Interruptions

Night wakings (bathroom breaks, discomfort)

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Environmental factors affecting sleep (noise, lighting)

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Frequent bathroom trips are the most common interruptions

Anabhayan

QUANTITATIVE ANALYSIS

Sleep Timing and Duration

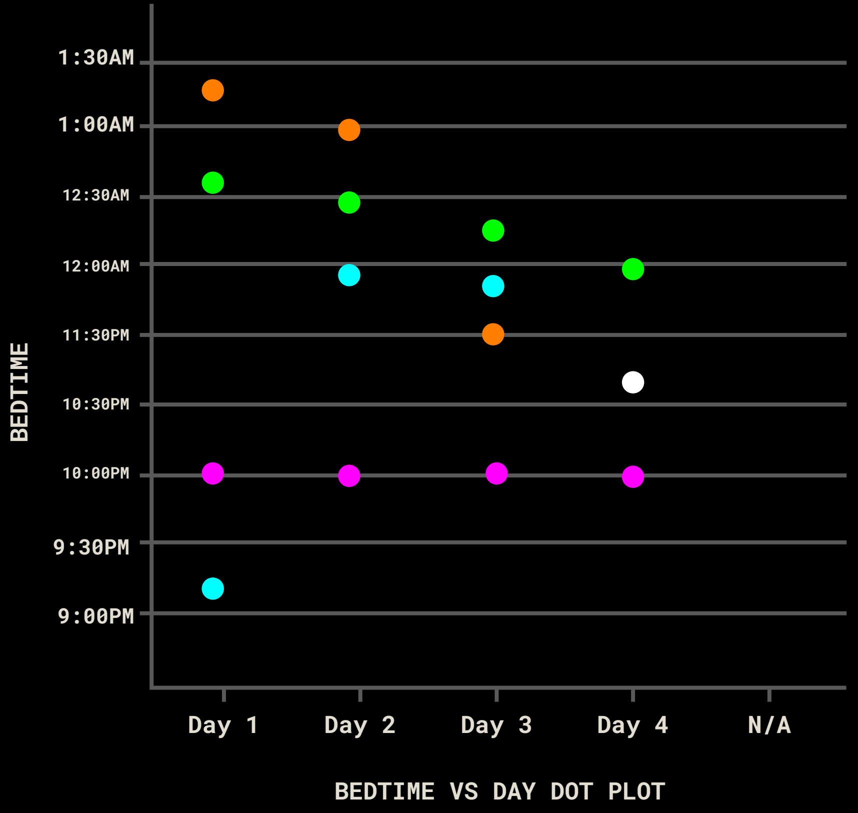
- ★ Average Bedtime: 12:30 am
- ★ Earliest Bedtime: 9:30 pm
- ★ Latest Bedtime: 1:20 am
- ★ Sleep Duration Range: 6 - 10.5 hours
- ★ Nights with Disrupted Sleep: 25 - 50%, depending on the participant

Mood & Alertness

- ★ Mood after completing nighttime routine was consistently higher than their starting mood before completing their nighttime routine.
- ★ Morning mood and preparedness ratings were higher when nighttime routines were completed the night prior.
- ★ Lower sleep quality was associated with, physical discomfort, frequent sleep disruptions throughout the night, and short sleep duration.

Example Group Trends

- ★ Completing a routine = higher mood ratings before bed
- ★ Longer sleep duration = higher alertness, focus, and mood next morning
- ★ Physical sleep disruptions = lower sleep quality regardless of nighttime routine completion



Themes

Group 1

eme 1: Consistent Sleep Improves Mood and Productivity

Section 1

Sub-Themes:

- Maintaining 7–9 hours of sleep improves energy, focus, and alertness
- Regular bedtime stabilizes mood in the morning

Description:
Participants who maintain a consistent sleep schedule report higher mood, alertness, and performance throughout the day. Quantitative data shows positive correlations between sleep duration and morning mood scores.

Group 2

heme 2: Irregular or Short Sleep Reduces Energy and Mood

Section 2

Sub-Themes:

- Less than 7 hours sleep → lower energy and focus
- Irregular bedtime → increased fatigue, anxiety, or stress

Description:
Short or inconsistent sleep schedules negatively affect morning mood, energy, and cognitive function, as confirmed by quantitative ratings and qualitative reflections.

Group 3

me 3: Nighttime Routines Support Sleep Quality

Section 3

Sub-Themes:

- Hygiene routines help mentally prepare for sleep
- Reading, journaling, meditation, or music/podcasts support relaxation
- Limited phone/social media before bed enhances sleep onset

Description:
Structured nighttime routines improve mood after the routine and next-morning alertness. Emotional and mental wind-down activities correlate with higher mood ratings.

Group 4

eme 4: Night Wakings Slightly Disrupt Sleep

Section 4

Sub-Themes:

- Frequent bathroom trips and minor disruptions are common
- Sleep quality is slightly affected, but routines mitigate negative effects

Description:
Minor night wakings do not fully prevent the benefits of a consistent nighttime routine but slightly reduce sleep efficiency.

Group 5

Theme 5: Late-Night gnitive Load and Screen Use Can Hinder Sleep

Section 1

Sub-Themes:

- Studying late or excessive phone use → later sleep onset and lower sleep quality
- Awareness of screen use can motivate participants to limit it before bed

Description:
Late-night cognitive activity and technology use negatively impact mood, sleep quality, and morning alertness. Participants often report better outcomes when limiting these behaviors.

https://www.figma.com/board/fD5eHXObbtWvypnRbBLVNg/Sleep-Diary-Study-%E2%80%93-Group-Affinity-Diagram?t=IFwe38fFbGcgUMG2-1

Yasmin Carter

“When I get enough sleep, I feel more focused and motivated the next day but staying consistent is the hardest part.”

ATTRIBUTES

- Health conscious but inconsistent
- Motivated by productivity
- Self-aware of sleep habits
- Tech dependent at night

Age	20
Occupation	College student
Status	Single
Education	Electrical engineer
Location	Newark NJ
Archetype	The Striving Student

BIO

Yasmin is a college student balancing coursework, studying, and personal time. Her sleep schedule often shifts depending on academic workload and late night activities. While she understands the importance of good sleep, maintaining consistent habits is challenging. Her mood, motivation, and productivity are closely tied to how well she sleeps the night before.

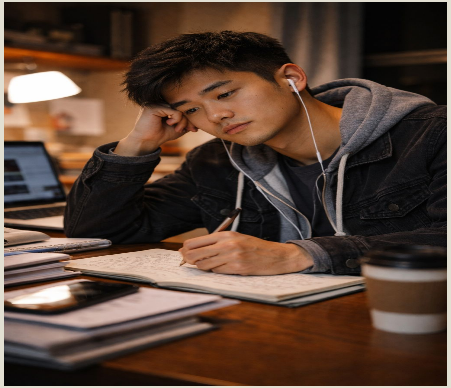
GOALS & NEEDS

- Maintain a consistent sleep schedule Yasmin wants to go to bed and wake up at consistent times so she can feel balanced and less tired throughout the day.
- Improve daytime focus and productivity She wants her sleep to support her academic performance and energy levels during classes and studying.
- Create a calming nighttime routine Yasmin needs simple wind-down activities that help her relax mentally and fall asleep faster.

FRUSTRATIONS

- Late-night studying disrupts sleep Academic demands often push her bedtime later than planned.
- Phone use before bed delays sleep onset Scrolling on social media makes it harder for her to wind down.
- Inconsistent sleep leads to low energy and mood Short or irregular sleep negatively affects her mornings.

Jordan Park



“I know sleep matters, but I usually sacrifice it to get everything done.”

Age	21
Occupation	College student
Status	Single
Education	Electrical engineer
Location	Newark NJ
Archetype	The Overloaded Achiever

BIO

Jordan is a college student who frequently stays up late studying or completing assignments. Unlike students who aim for routine, Jordan prioritizes deadlines over sleep consistency. While they can manage short term fatigue, extended sleep loss leads to anxiety, low mood, and difficulty concentrating.

ATTRIBUTES

- Highly driven under pressure
- Deadline focused
- Sleep deprived but resilient
- Aware of consequences, ignores them short term

GOALS & NEEDS

- **Maximize productivity despite limited sleep**
  Jordan wants to stay functional even when sleep is short due to academic demands.
- **Reduce fatigue without changing schedule drastically**
  They want small improvements that fit into an already busy routine.
- **Avoid burnout during heavy study periods**
  Jordan needs strategies to protect mood and focus during stressful weeks.

FRUSTRATIONS

- **Late night studying cuts into sleep time**
  Academic pressure makes it hard to stop working.
- **Irregular bedtimes cause morning exhaustion**
  Jordan often wakes up feeling unprepared for the day.
- **Mental fatigue affects focus and motivation**
  Lack of sleep reduces efficiency, even when effort is high.

Design Opportunity 1: Bridge the Gap Between Screen-Time Awareness & Action

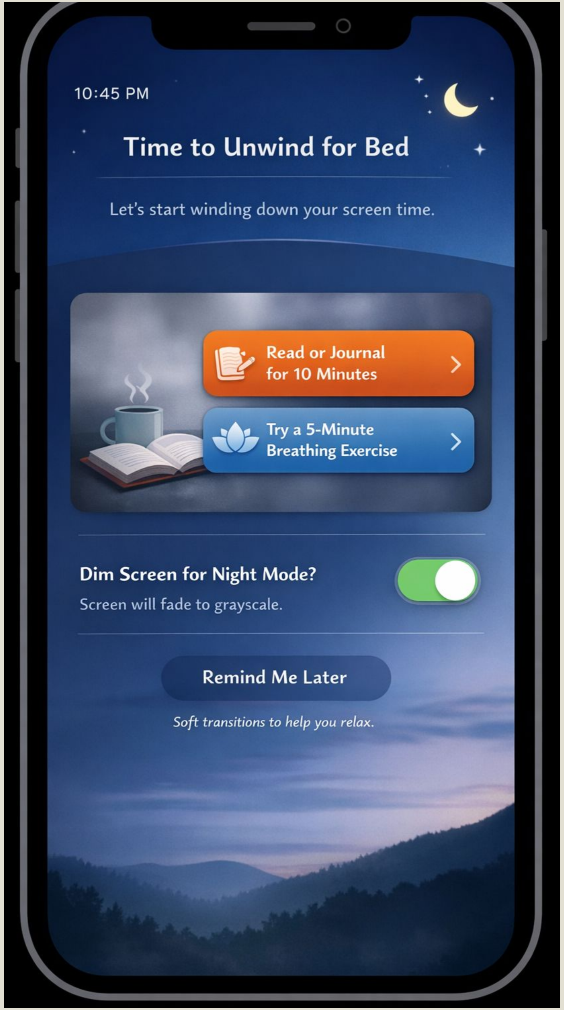
Most college students already know screens affect sleep. Our study shows the problem isn't awareness, it's follow-through. Late at night, students are cognitively depleted, stressed, and socially pulled toward their phones.

Design Ideas:

- ★ Gradually grayscale or dim apps during wind-down time instead of hard cutoffs.
- ★ Offer replacement behaviors such as reading prompts, breathing exercises

Benefits Students Will Experience:

- ★ Reduced stimulation before sleep
- ★ Stronger association between night-time and rest
- ★ Increased confidence in managing habits without feeling controlled



Design Opportunity 2: Emotional Decompression Before Bed

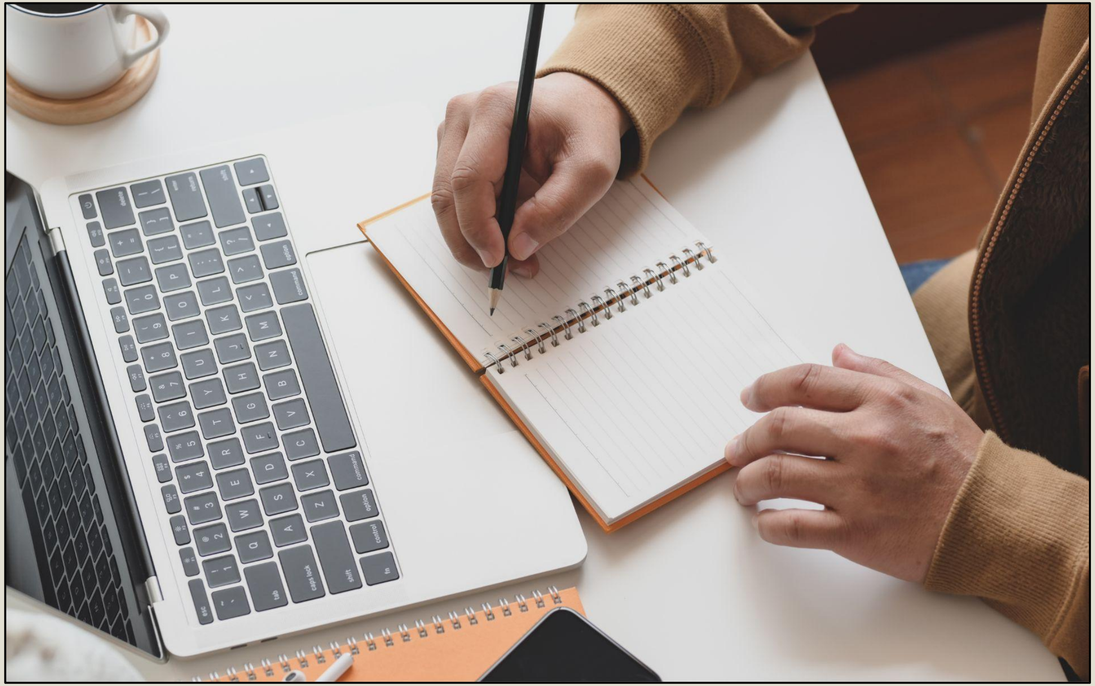
Our diary study shows emotional decompression consistently improves mood before sleep. College students carry unresolved stress from academics, work, and social life into bed, which keeps the mind active even when the body is tired.

Design Ideas:

- ★ One-minute decompression prompts for example: *“What part of today are you ready to leave behind?”*
- ★ Avoiding metrics and emphasize emotional release rather than tracking streaks.

Benefits Students Will Experience:

- ★ Better emotional regulation
- ★ Reduced anxiety and racing thoughts
- ★ Improved morning mood and alertness



Follow-up Study Plan

If we had the opportunity we would conduct a follow up study to see if our recommendations would be effective at honing in college student sleep schedules. The study would be formatted as a semi structured interview study in order ask them directly about:

★ Whether or not our solutions worked
★ What caused them to succeed and fail
★ What did their nightly routines look like before and after incorporating our recommendations.

Research Method

★ Semi-Structured Interview

Participants

★ 10-15 college students

Data Collection

★ Interviews would include guiding questions, open-ended questions, and some multiple choice questions
★ 15-30 minute interviews
★ We would focus mostly on any barriers in regards to good quality sleep and screen time before bed time, motivations in regards to productivity and nighttime routines, and different contexts

Conclusion

Key Takeaways

★ Activities included in nighttime routines and nighttime routine consistency have a strong impact on sleep quality and mood the next morning
★ Emotional and physical factors both play important roles in sleep
★ Structural support is needed to help people change their habits

Impact

★ The data collected gave us great insights that can guide design improvements and creation of student wellness tools to support students in building better habits to promote better sleep and performance for students

THANK YOU

"Sleep tight and don't let the bedbugs bite!"

- *Your Mom*